

Pulmonary Circulation, Pulmonary Edema, Pleural Fluid

Pulmonary Edema

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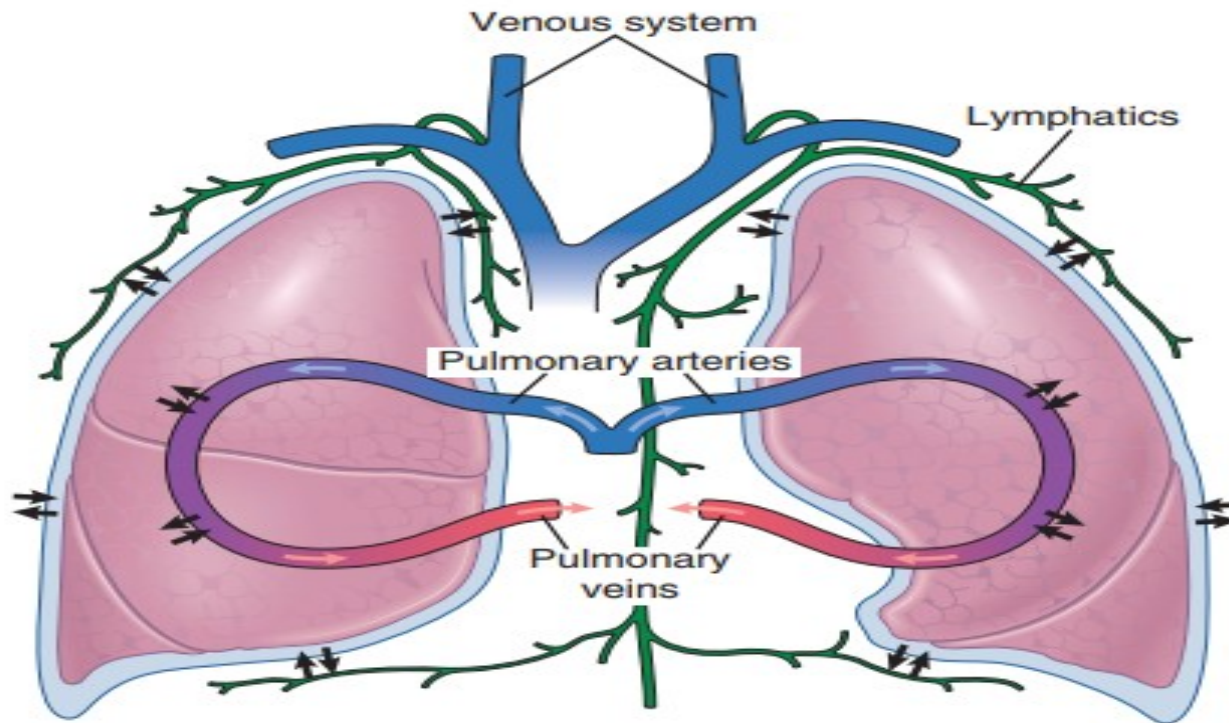
- Pulmonary edema occurs in the same way that edema occurs elsewhere in the body.
- Any factor that **increases fluid filtration** out of the pulmonary capillaries or that impedes pulmonary lymphatic function and causes the pulmonary **interstitial fluid pressure to rise** from the negative range into the positive range will cause rapid filling of the pulmonary interstitial spaces and alveoli with large amounts of free fluid.

Cont...

- The most **common causes** of pulmonary edema are as follows:
 1. Left-sided **heart failure or mitral valve disease**, with consequent great increases in pulmonary venous pressure and pulmonary capillary pressure and flooding of the interstitial spaces and alveoli.
 2. Damage to the pulmonary blood capillary membranes caused by **infections such as pneumonia** or by **breathing noxious substances** such as chlorine gas or sulfur dioxide gas.
- Each of these mechanisms causes rapid leakage of both plasma proteins and fluid out of the capillaries and into both the lung interstitial spaces and the alveoli

FLUID IN THE PLEURAL CAVITY

- When the lungs expand and contract during normal breathing, they slide back and forth within the pleural cavity.
- To facilitate this movement, a thin layer of mucoid fluid lies between the parietal and visceral pleurae.



Cont...

- The total amount of fluid in each pleural cavity is normally slight—only a few milliliters.
- Whenever the quantity becomes more than barely enough to begin flowing in the pleural cavity, **the excess fluid is pumped away by lymphatic vessels** opening directly from the pleural cavity into (1) the mediastinum, (2) the superior surface of the diaphragm, and (3) the lateral surfaces of the parietalpleura.
- Therefore, the *pleural space*—the space between the parietal and visceral pleurae—is called a ***potential space*** because it normally is so narrow that it is not obviously a physical space.

Pleural Effusion

- **Pleural Effusion—Collection of Large Amounts of Free Fluid in the Pleural Space.**
- Pleural effusion is analogous to edema fluid in the tissues and can be called “**edema of the pleural cavity.**”

